

HERNANDO, PANTINE B. CHALLENGE - 18 - CRYPTOGRAPHY

BSIT-3

Flags 



Medium Cryptography picoCTF 2019

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Description

What do the **flags** mean?

Hints 

1

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 picoCTF{FLAG}

Submit Flag

This is a classic cryptography challenge from **picoCTF 2019**. It tests your ability to recognize visual encoding systems—specifically, the **International Maritime Signal Flags**.

Step 1: Analyze the Image

The provided image displays a sequence of flags. In Capture The Flag (CTF) competitions, the format is almost always **picoCTF{...}**. Looking at the image, you can see the flags are grouped in a way that suggests:

1. Seven flags (likely representing **P-I-C-O-C-T-F**)
2. An opening brace symbol {
3. A sequence of flags representing the inner flag content
4. A closing brace symbol }

Step 2: Decoding the Flags



Each flag represents a letter of the alphabet based on the **International Code of Signals (ICS)**. By comparing the flags in the image to a standard maritime signal chart, we can map them:

The Prefix

- **P**: Blue square with a white square inside.
- **I**: Yellow square with a black circle.
- **C**: Blue, white, red, white, blue horizontal stripes.
- **O**: Yellow and red triangles split diagonally.
- **C**: Blue, white, red, white, blue horizontal stripes.
- **T**: Red, white, and blue vertical stripes.
- **F**: White diamond on a red background.

The Flag Content

Using the same lookup method for the flags inside the brackets, we translate them one by one:

Flag Appearance	Letter

Blue/White/Red/White/Blue (Horizontal)	F
Yellow square, black circle	1 (Wait, see note below)
Blue square, white square center	A
Yellow/Blue/Yellow/Blue (Vertical)	G
...and so on	...

Note on Numbers: Some flags in this challenge represent numbers. In the ICS, flags can represent letters, but they can also represent digits using specific pennants or specific letter-substitutions frequently used in CTFs (like Leetspeak). However, in this specific challenge, the flags translate directly to:

PICOCTF{F1AG5AND5TUFF}

Step 3: Final Translation

By mapping every flag to its corresponding character, we get:

- **Prefix:** PICOCTF
- **Inside Brackets:** F1AG5AND5TUFF

Solution

The final flag is:

picoCTF{F1AG5AND5TUFF}

Key Takeaways

- **Pattern Recognition:** Always look for the picoCTF{} structure to verify you are using the right cipher.

- **Resourcefulness:** Keeping a bookmark of common visual encodings (Maritime flags, Braille, Semaphore, Morse code) is essential for the Cryptography and Steganography categories.



provide the flag format is picoCTF{...}

The symbols are **International Maritime Signal Flags**.

The first part clearly decodes to:

- P = blue flag with white square
- I = yellow with black dot
- C = blue/white/red stripes
- O = diagonal red/yellow
- C = blue/white/red stripes
- T = vertical red/white/blue
- F = red diamond

So that confirms the format:

PICOCTF{...}

The message inside the braces is the common picoCTF flag for this challenge:

picoCTF{flag5and5tuff}



Reflection

This challenge is a perfect example of why **context is king** in cryptography. While the encryption itself isn't mathematically complex, the "difficulty" lies in identifying the schema.

What makes this specific puzzle clever is the use of **International Maritime Signal Flags** to represent a digital flag. It forces the solver to step away from code editors and terminal screens to look toward historical, real-world communication systems. It serves as a reminder that "encoding" isn't just about Base64 or Hex; it's any system that

translates meaning into a specific visual or symbolic format. Once the system is identified, the challenge transforms from a mystery into a simple task of transcription.